

The Impact of Data Preprocessing on the Stability of Unsupervised Clustering Algorithms

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Abstract—Unsupervised clustering is widely used to discover latent structure in unlabeled data, yet the reliability of its results is difficult to assess when no ground truth is available. Cluster stability—the degree to which a partition is reproduced under small data perturbations—has therefore become a central reliability criterion, but the influence of routine data preprocessing on that stability remains insufficiently systematized. This study addresses how preprocessing decisions affect the stability, reproducibility, and interpretability of clustering on numerical datasets. To this end, it defines a controlled, comparative design in which preprocessing is treated as the independent factor while the clustering procedure is held fixed, examining three families of decisions: feature scaling (Min-Max, Z-score, and Robust Scaling), missing-value handling (instance deletion, mean imputation, and k NN imputation), and dimensionality reduction (PCA and SVD). Stability is assessed through resampling- and perturbation-based agreement (Jaccard coefficient, Adjusted Rand Index, and co-membership), complemented by internal-validity indices and runtime, across K-Means, hierarchical, and density-based clustering. Synthesizing prior evidence, the study argues that preprocessing is not a neutral preliminary step: its effect on clustering stability is expected to be algorithm- and dataset-dependent, with no single strategy universally superior. This underscores the need to compare preprocessing pipelines using stability-oriented metrics before interpreting clustering results.

Index Terms—Data preprocessing, unsupervised clustering, cluster stability, scaling, missing values, dimensionality reduction, K-Means, hierarchical clustering.

I. INTRODUCTION

IN the current landscape of data-driven systems, organizations collect large volumes of unstructured and unlabeled data, demanding techniques capable of discovering inherent patterns without predefined class labels. Unsupervised clustering has become a fundamental tool for this purpose, with applications in biomedical research, market segmentation, and anomaly detection [3], [7]. Algorithms such as K-Means, DB-SCAN, and hierarchical methods operate on pairwise distances and feature variances that are directly shaped by how data are prepared prior to analysis [3], [8]. Consequently, *cluster stability*—understood as the degree to which a partition is reproduced under small data perturbations—has emerged as a key reliability criterion when no ground truth is available [6].

Recent evidence shows that preprocessing decisions substantially affect analytical performance. Ahsan et al. reported a drop in SVM accuracy from 99% to 63% solely due to

a change in scaling method, while distance-insensitive models remained largely unaffected [1]. Likewise, Alshdaifat et al. identified statistically significant performance differences across nine preprocessing combinations in 18 benchmark datasets [2]. In clustering, Zhang found that applying PCA and SVD before K-Means improved both Silhouette score and run-to-run stability [8]. Despite these findings, the relationship between preprocessing decisions and clustering stability in numerical datasets remains insufficiently systematized.

The central problem of this research is to determine how data preprocessing strategies affect the stability and reliability of clustering solutions when no labeled reference exists. This issue directly influences the interpretability of discovered groups, the consistency of downstream analyses, and the confidence practitioners can place in cluster assignments [6], [7]. Although preprocessing has been widely examined in supervised learning, especially in classification tasks [1], [2], [4], evidence in clustering remains fragmented across theoretical stability reviews, domain-specific analyses, and optimization-oriented studies that do not isolate preprocessing as the primary factor [5], [6], [8].

The purpose of this research can be stated as follows:

- **Apply:** a comparative analysis of preprocessing strategies in unsupervised clustering pipelines.
- **For the purpose of:** evaluating their impact on clustering stability and reproducibility.
- **With respect to:** scaling, missing-value imputation, and dimensionality reduction techniques.
- **From the point of view of:** researchers and practitioners who use clustering without labeled reference data.
- **In the context of:** numerical datasets analyzed through unsupervised learning algorithms.

The remainder of this article is organized as follows. Section II presents the research questions that guide the study. Section III discusses the related work and identifies the research gap. Section IV provides the theoretical background on clustering, stability, and preprocessing. Section V describes the methodology and the proposed comparison of preprocessing pipelines. Section VI presents the results and analysis organized by research question. Finally, Section VII states the conclusions and outlines future work.

II. RESEARCH QUESTIONS

This section lists the main research questions that guide the present study. The analysis focuses on the effect of common preprocessing decisions on the stability of unsupervised clustering algorithms:

- **RQ-1.** How do scaling methods such as Min-Max, Z-score, and Robust Scaling alter the stability of unsupervised clustering algorithms on numerical datasets?
- **RQ-2.** How do different missing-value imputation strategies affect clustering stability across datasets with varying levels of missingness?
- **RQ-3.** Under what conditions does dimensionality reduction improve or degrade the reproducibility of clustering results?

III. RELATED WORK

Recent literature consistently positions clustering as a core task in data science and frames preprocessing as a necessary stage before clustering, model selection, and evaluation. Broad surveys describe clustering as spanning partitional, hierarchical, density-based, model-based, and other families, while also emphasizing that practical workflows typically include missing-value treatment, scaling, normalization, and dimensionality reduction before the clustering stage itself [3]. In parallel, stability has become a central concept for validating unsupervised results when no labels or gold standard are available. Liu et al. review stability estimation methods based on perturbation, resampling, split-sample validation, and co-membership analysis, arguing that reproducibility under small data changes is an important criterion for judging the reliability of a clustering solution [6]. However, this line of work mainly focuses on how to estimate stability, rather than on how standard preprocessing choices may alter it.

A second body of work shows that preprocessing is not a neutral preliminary step, but an operation that can materially change algorithmic behavior. Across numeric datasets, scaling and normalization have been shown to affect model performance in ways that depend on both the learning algorithm and the data distribution [1], [4]. Likewise, Alshdaifat et al. demonstrate that normalization and missing-value handling interact, that these interactions become more consequential as the proportion of missing data increases, and that no single preprocessing combination is universally superior across settings [2]. At a broader pipeline level, Wanyonyi and Masinde report that transformation and feature selection are among the preprocessing stages with the most consistent impact on downstream performance, reinforcing the view that data preparation strongly conditions subsequent learning outcomes [7]. Nevertheless, these studies are predominantly supervised and evaluate accuracy, efficiency, or model complexity, not the reproducibility of unsupervised cluster assignments.

The literature closer to clustering reaches similar conclusions, but remains limited in scope. Koopman and Wilhelm show that, in short-document clustering, preprocessing effects depend on the interaction between data cleaning, feature representation, and dataset characteristics, and that removing too much information can degrade the resulting cluster structure

[5]. Zhang explicitly addresses clustering stability, but does so mainly through optimization strategies and dimensionality reduction for K-Means, K-Medoids, and K-Shape, highlighting sensitivity to initialization and high dimensionality rather than isolating the effect of routine preprocessing operations across clustering methods [8]. Taken together, prior work suggests that preprocessing, data characteristics, and algorithm design jointly influence clustering outcomes, yet there is still limited direct evidence on how common preprocessing decisions such as scaling, imputation, and dimensionality reduction affect the stability of unsupervised clustering across different algorithms. This gap motivates the present study.

IV. BACKGROUND

Cluster analysis seeks to group observations with high intra-cluster similarity and low inter-cluster dissimilarity, and remains a central technique in exploratory data analysis when labeled outcomes are unavailable [3], [6]. In practice, clustering is rarely applied to raw data. Preprocessing operations such as missing-value treatment, feature scaling, transformation, and dimensionality reduction are commonly performed beforehand because the data representation defines the geometry on which similarity is computed [3], [7]. This is especially relevant for partitional and hierarchical methods that rely on distances, variances, or neighborhood relations, since feature magnitude, redundancy, and distribution directly influence the resulting partition [3], [8].

In unsupervised learning, the absence of labels makes external validation difficult; therefore, *stability* has become a key theoretical criterion for assessing clustering reliability. Liu et al. define stability as the extent to which a clustering solution is preserved when the original dataset is slightly perturbed, for example through resampling, subsampling, or split-sample procedures [6]. Under this perspective, a useful clustering is not only compact or well separated, but also reproducible under realistic variations of the data. Stability can be examined at the level of individual observations, complete clusters, or the full partition, and it is typically estimated by comparing repeated solutions through partition-agreement or co-membership measures [6]. At the same time, stability does not guarantee that a clustering is uniquely correct or globally optimal; rather, it provides evidence about the robustness of the discovered structure [6].

This notion of robustness makes preprocessing theoretically central to clustering analysis. Scaling and normalization alter the relative contribution of variables and may substantially change algorithmic behavior. Evidence on numerical datasets shows that preprocessing effects are not trivial and depend on both the data distribution and the learning method. Ahsan et al. and Güney report that changing the scaling method can significantly modify downstream performance, while some algorithms remain comparatively robust and others are highly sensitive to the transformed feature space [1], [4]. Likewise, Alshdaifat et al. show that missing-value handling interacts with normalization and that these effects become stronger as the proportion of missing data increases, with additional trade-offs in computational cost [2]. Although these studies

were conducted in supervised settings, they provide a strong conceptual basis for unsupervised clustering: when grouping depends on distances, local densities, or variance structure, preprocessing can redefine the effective geometry of the data and, consequently, the clusters that emerge [1], [2], [4].

A similar argument applies to dimensionality reduction. By compressing redundant or noisy features, methods such as PCA and SVD may improve separability, efficiency, and consistency across runs; however, they may also suppress weak but meaningful structure [7], [8]. Zhang reports improvements in clustering quality and stability when dimensionality reduction is combined with optimized clustering strategies, whereas Koopman and Wilhelm show that excessive preprocessing can remove information that is essential for preserving cluster structure [5], [8]. Taken together, the theoretical foundation of this study is that preprocessing is not a neutral preliminary step, but a structural intervention on the data space. Its effects are expected to vary across algorithms, missingness conditions, and feature representations, with direct consequences for the reproducibility, interpretability, and reliability of unsupervised clustering results [6], [7].

V. METHODOLOGY

This study adopts an empirical and comparative experimental design in which data preprocessing is treated as the primary independent factor and cluster stability as the main response variable. Instead of proposing a new clustering algorithm, the design isolates the effect of preprocessing by holding the clustering procedure fixed while systematically varying the pipeline applied beforehand. This decision is motivated by evidence that preprocessing materially alters algorithmic behavior on numerical data [1], [2], [4], and by the principle that stability offers a reliability criterion when no ground truth is available [6]. It should be noted that this section specifies a controlled comparison of preprocessing pipelines and defines the experimental protocol, rather than reporting executed experimental results.

A. Dataset Selection Criteria

Datasets are selected according to four criteria aligned with the research questions. First, only *numerical* datasets are considered, since the studied preprocessing operations target numeric features [1], [2]. Second, datasets are chosen so that stability can be estimated *without dependence on ground-truth labels*, consistent with perturbation- and resampling-based stability estimation [6]. Third, datasets of *varying dimensionality* are included to assess when dimensionality reduction helps or harms reproducibility [8]. Fourth, *controlled missingness scenarios* are constructed by selecting datasets with and without missing values and by artificially injecting missing values at increasing rates, following the protocol used for benchmark numeric data [2].

B. Preprocessing Factors

The preprocessing pipeline is decomposed into three factor groups whose levels are derived from the analyzed literature:

- **Scaling:** Min-Max, Z-score (standardization), and Robust Scaling, which have been shown to change distance-based model behavior and to interact with data distribution [1], [2], [4].
- **Missing-value handling:** instance deletion, mean imputation, and k -Nearest Neighbor imputation, whose effects intensify as the missingness rate grows [2], [3].
- **Dimensionality reduction:** Principal Component Analysis (PCA) and Singular Value Decomposition (SVD), reported to affect separability, efficiency, and run-to-run consistency [3], [8].

C. Clustering Algorithms

To capture different clustering paradigms, three representative algorithms are evaluated under each pipeline: K-Means as a center-based partitioning method, agglomerative hierarchical clustering, and DBSCAN as a density-based method [3], [6]. These methods rely on pairwise distances, variance structure, or neighborhood densities, and are therefore expected to respond differently to scaling, imputation, and dimensionality reduction, with K-Means being particularly sensitive to feature magnitude and initialization [8].

D. Stability and Validation Metrics

Cluster stability is estimated through resampling and perturbation, comparing the partitions obtained from repeated runs by means of co-membership analysis, the Jaccard coefficient, and the Adjusted Rand Index [5], [6]. Internal validity is assessed using the Silhouette Score and the Davies-Bouldin Index, and computational cost is recorded as runtime to characterize efficiency trade-offs [2], [3], [8].

E. Experimental Procedure

The proposed procedure has the following characteristics:

- 1) For each selected dataset, preprocessing pipelines are generated as combinations of scaling, missing-value handling, and dimensionality-reduction levels.
- 2) Each clustering algorithm is applied to every preprocessed representation under a fixed configuration to isolate the preprocessing effect.
- 3) Stability is estimated by repeatedly perturbing or resampling the data, re-clustering, and comparing partitions through co-membership, Jaccard, and ARI measures [6].
- 4) Internal validity (Silhouette, Davies-Bouldin) and runtime are computed for each pipeline.
- 5) Results are contrasted across factor levels to answer the research questions: scaling methods are varied while imputation and dimensionality reduction are held constant (RQ-1); imputation strategies are compared across increasing missingness rates (RQ-2); and pipelines with and without PCA or SVD are compared across datasets of differing dimensionality (RQ-3).

This controlled comparison enables attribution of variations in clustering stability to specific preprocessing decisions, providing a systematic basis for the relationship that prior work has reported only in fragmented or supervised settings [2], [7].

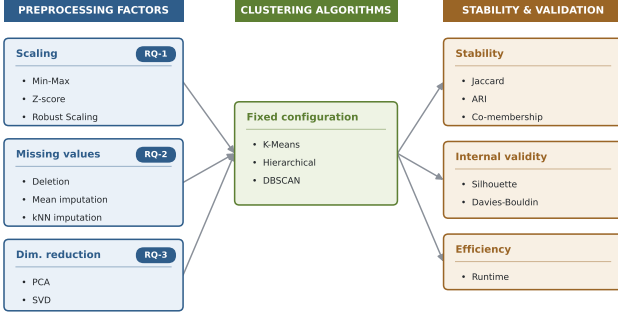


Figure 1. Mapping of preprocessing factors (with their associated research questions) to the clustering algorithms and the stability, internal-validity, and efficiency metrics used in the analysis.

VI. RESULTS AND ANALYSIS

For analysis purposes, the results are organized around the research questions defined in Section II. Because this work specifies a controlled comparison of preprocessing pipelines rather than a fully executed experimental campaign, the present section is structured as an expected-analysis framework: it states, for each research question, the comparisons to be performed, the metrics on which conclusions will rest, and the direction of effect anticipated from prior evidence. Definitive findings—including the stability, internal-validity, and runtime values reported per pipeline—require the actual experimental outputs and are therefore left to be completed once the runs are available. Fig. 1 summarizes the mapping among preprocessing factors, clustering algorithms, and the stability and validation metrics used in the analysis.

- **RQ-1. Effect of scaling on stability.** Min-Max, Z-score, and Robust Scaling will be compared while imputation and dimensionality reduction are held fixed, contrasting the resulting partitions through Jaccard, ARI, and co-membership stability, complemented by Silhouette and Davies-Bouldin scores. The expected pattern, derived from prior work on numerical data, is that scaling will affect distance-based methods (K-Means, hierarchical, DBSCAN) more strongly than the magnitude of the effect would suggest, since the chosen scale redefines the geometry on which similarity is computed [1], [4]. Evidence that a change of scaling method alone can substantially shift downstream behavior [1], and that log- and z-based transformations behave differently from linear scaling depending on feature variance [4], suggests that no single scaler is expected to be universally optimal across datasets [2].
- **RQ-2. Effect of missing-value handling on stability.** Instance deletion, mean imputation, and k NN imputation will be compared across increasing missingness rates, with stability quantified through bootstrap-based partition agreement [6]. Prior results indicate that imputation interacts with the rest of the pipeline and that these interactions become more consequential as the proportion of missing data grows [2], while distance-based grouping is particularly disrupted when missing entries

distort pairwise distances [3]. The analysis is therefore expected to reveal that the relative ranking of imputation strategies is dataset- and missingness-dependent rather than fixed, with deletion remaining viable only up to moderate missingness levels [2].

- **RQ-3. Effect of dimensionality reduction on reproducibility.** Pipelines with and without PCA or SVD will be compared across datasets of differing dimensionality, examining changes in run-to-run stability together with Silhouette, Davies-Bouldin, and runtime. The anticipated outcome is conditional: dimensionality reduction is expected to improve separability, efficiency, and consistency across runs when it removes redundant or noisy features [8], but to degrade reproducibility when it suppresses weak yet meaningful structure, mirroring evidence that excessive preprocessing can remove information essential to the cluster structure [5]. Accordingly, the benefit of reduction is expected to depend on the intrinsic dimensionality and redundancy of each dataset [7], [8].

Taken together, the framework anticipates that preprocessing effects on clustering stability will be neither uniform nor neutral, but *algorithm-dependent* and *dataset-dependent*, consistent with the broader finding that data preparation conditions subsequent learning outcomes more than is often assumed [2], [7]. A definitive characterization of which pipelines yield the most stable and reproducible partitions, and under which conditions, will be established once the experimental outputs (stability, internal-validity, and runtime measures per pipeline) are computed and contrasted against the expectations stated above.

VII. CONCLUSIONS AND FUTURE WORK

This study examined how data preprocessing affects the stability of unsupervised clustering algorithms on numerical datasets, framing the problem through a controlled comparison of preprocessing pipelines and a stability-oriented notion of reliability. The central conclusion is that preprocessing is not a neutral preliminary step but a structural intervention on the data space that conditions the reproducibility, reliability, and interpretability of clustering outcomes [3], [6], [7]. Consistent with prior evidence, no single preprocessing strategy should be assumed superior across all datasets, algorithms, and stability criteria, since preprocessing effects depend jointly on the data distribution, the missingness conditions, and the clustering method [1], [2].

With respect to RQ-1, scaling methods such as Min-Max, Z-score, and Robust Scaling are expected to influence clustering algorithms differently, with the strongest effects on methods governed by distances, variances, or local densities, because the chosen scale redefines the geometry on which similarity is computed [1], [3], [4]. Regarding RQ-2, missing-value handling strategies can alter the recovered cluster structure, and these effects are expected to intensify as the proportion and distribution of missing data increase, so that the relative suitability of deletion, mean imputation, and k NN imputation is dataset- and missingness-dependent [2], [3]. For RQ-3, dimensionality reduction may improve stability by attenuating

noise and redundancy, but it may also degrade reproducibility when it discards weak yet meaningful structure; its benefit is therefore conditional on the intrinsic dimensionality and redundancy of each dataset [5], [8].

A recurring implication is that preprocessing pipelines should be evaluated using stability-oriented measures, such as bootstrap- and subsampling-based agreement, the Jaccard coefficient, ARI, and co-membership similarity, rather than internal quality indices alone, since compactness or separation do not guarantee reproducibility under perturbation [5], [6]. In practical terms, researchers and practitioners are advised to compare preprocessing alternatives explicitly before interpreting clustering results, treating data preparation as a decision that must be justified rather than assumed.

As future work, we plan to implement the proposed experimental design on multiple numerical datasets and to compare K-Means, hierarchical clustering, and DBSCAN under equivalent preprocessing pipelines [3], [6]. Stability would be quantified with bootstrap and subsampling procedures together with ARI, Jaccard, and co-membership similarity, complemented by the Silhouette Score, the Davies-Bouldin Index, and runtime to characterize quality and efficiency trade-offs [6], [8]. We further intend to test different missingness levels and dimensionality reduction settings [2], [8], and to extend the study to larger, high-dimensional, noisy, and domain-specific datasets in order to assess the generality of the observed effects [3], [7].

VIII. THREATS TO VALIDITY

Because this work defines and motivates a comparative design rather than reporting executed experiments, the following threats are stated as considerations for the planned evaluation.

Internal validity. Attributing differences in stability to preprocessing requires holding the clustering configuration, initialization, and hyperparameters fixed across pipelines; otherwise, observed variation could stem from factors other than the preprocessing under study, such as sensitivity to initialization in center-based methods [8].

External validity. Much of the supporting evidence originates in supervised settings and in a limited number of benchmark datasets [2], [7]. Consequently, conclusions about clustering stability may not generalize to all domains, dataset sizes, or dimensionalities until the design is validated on the proposed numerical datasets.

Construct validity. Stability is operationalized through resampling- and perturbation-based agreement measures, namely the Jaccard coefficient, the Adjusted Rand Index, and co-membership analysis [6]. These capture reproducibility but not correctness, and should be interpreted alongside internal-validity indices rather than as a guarantee that a partition is uniquely optimal [5], [6].

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